

DBLP is a computer science bibliography website. Starting in 1993 at Universität Trier in Germany, it grew from a small collection of HTML files and became an organization hosting a database and logic programming bibliography site. DBLP listed more than 5.4 million journal articles, conference papers, and other publications on computer science in December 2020, up from about 14,000 in 1995 and 3.66 million in July 2016. In this notebook I'm retrieving data from dblp api and making various visualizations.

```
import requests
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
```

1. Creating the Data retrieval function

I'm creating a function that retrieves the title of each publication, the year and the type

```
def get_data(author_name):
    titles=[]
    years=[]
    types=[]
    base_url="https://dblp.org/search/publ/api"
    params={"q":author_name,"format":"json"}
    response = requests.get(base_url, params=params)
    if response.status_code == 200:
        data = response.json()
        publications = data.get('result', {}).get('hits',
{}).get('hit', [])
        for pub in publications:
            info = pub.get('info', {})
            title = info.get('title')
            year = info.get('year')
            book_type=info.get("type")
            titles.append(title)
            years.append(year)
            types.append(book_type)
        d={"Title":titles,"Year":years,"Type":types}
        d=pd.DataFrame(data=d)
        return d
```

I am adding an extra column "Name", so I can separate publications by author later

```
vas_df=get_data("Vasileios Tenentes")
vas_df["Name"]="Vasileios Tenentes"
vas_df
```

	Title	Year	\
0	Multi-Vt-Based Energy Efficiency Optimization ...	2024	
1	RTL Flow for the Power Side-Channel Resilience...	2024	
2	Evaluating Trusted Firmware Remote Attestation...	2024	
3	High Throughput and Energy Efficient SHA-2 ASI...	2023	
4	SiCBit-PUF: Strong in-Cache Bitflip PUF Comput...	2023	
5	MetaSPICE: Metaprogramming SPICE Framework for...	2023	
6	Design and Evaluation of a Peripheral for Inte...	2023	
7	Embedded Platforms for Trusted Edge Computing ...	2023	
8	REVOLVER: A Zero-Step Execution Emulation Fram...	2022	
9	Energy Efficient Speech Command Recognition fo...	2021	
10	Analysis on Retention Time and Adaptive Refres...	2019	
11	Run-time Detection and Mitigation of Power-Noi...	2019	
12	Exploiting Aging Benefits for the Design of Re...	2018	
13	Leakage Current Analysis for Diagnosis of Brid...	2018	
14	Recycled IC detection through aging sensor.	2018	
15	Collective-Aware System-on-Chips for Dependabl...	2018	
16	Susceptible Workload Evaluation and Protection...	2017	
17	Aging Benefits in Nanometer CMOS Designs.	2017	
18	Coarse-Grained Online Monitoring of BTI Aging ...	2017	
19	Low cost error monitoring for improved maintai...	2017	
20	Hardware and software innovations in energy-ef...	2017	
21	Low power probabilistic online monitoring of s...	2017	
22	Online tuning of Dynamic Power Management for ...	2017	
23	The impact of BTI aging on the reliability of ...	2016	
24	Reliable Power Gating With NBTI Aging Benefits.	2016	
25	BTI aware thermal management for reliable DVFS...	2016	
26	Susceptible workload driven selective fault to...	2016	
27	DFT Architecture With Power-Distribution-Netwo...	2015	
28	NBTI and leakage aware sleep transistor design...	2015	
29	Diagnosis of power switches with power-distrib...	2015	

	Type	Name
0	Journal Articles	Vasileios Tenentes
1	Conference and Workshop Papers	Vasileios Tenentes
2	Conference and Workshop Papers	Vasileios Tenentes
3	Conference and Workshop Papers	Vasileios Tenentes
4	Conference and Workshop Papers	Vasileios Tenentes
5	Conference and Workshop Papers	Vasileios Tenentes
6	Conference and Workshop Papers	Vasileios Tenentes
7	Conference and Workshop Papers	Vasileios Tenentes
8	Conference and Workshop Papers	Vasileios Tenentes
9	Conference and Workshop Papers	Vasileios Tenentes
10	Conference and Workshop Papers	Vasileios Tenentes
11	Conference and Workshop Papers	Vasileios Tenentes
12	Journal Articles	Vasileios Tenentes
13	Journal Articles	Vasileios Tenentes
14	Conference and Workshop Papers	Vasileios Tenentes
15	Conference and Workshop Papers	Vasileios Tenentes
16	Journal Articles	Vasileios Tenentes

17	Journal Articles	Vasileios Tenentes
18	Journal Articles	Vasileios Tenentes
19	Conference and Workshop Papers	Vasileios Tenentes
20	Conference and Workshop Papers	Vasileios Tenentes
21	Conference and Workshop Papers	Vasileios Tenentes
22	Conference and Workshop Papers	Vasileios Tenentes
23	Journal Articles	Vasileios Tenentes
24	Journal Articles	Vasileios Tenentes
25	Conference and Workshop Papers	Vasileios Tenentes
26	Conference and Workshop Papers	Vasileios Tenentes
27	Journal Articles	Vasileios Tenentes
28	Conference and Workshop Papers	Vasileios Tenentes
29	Conference and Workshop Papers	Vasileios Tenentes

```
emin_df=get_data("Emin Borandag")
emin_df["Name"]="Emin Borandag"
emin_df
```

	Title	Year	\
0	Development of majority vote ensemble feature ...	2021	
1	Majority vote feature selection algorithm in s...	2019	
2	Vehicle plate tracking system.	2018	
3	TTC-3600: A new benchmark dataset for Turkish ...	2017	
4	Multi-level reranking approach for bug localiz...	2016	
5	A case study for the software size estimation ...	2016	
6	Regression Analysis Based Software Effort Esti...	2016	
7	The average scattering number of graphs.	2016	
8	Metin Madenciliği Kullanılarak Yazılım Kullanı...	2015	
9	Yazılım Hata Kestiriminde Kolektif Sınıflandır...	2015	
10	İlişkisel veri tabanı kullanılan yazılımlarda ...	2011	

	Type	Name
0	Journal Articles	Emin Borandag
1	Journal Articles	Emin Borandag
2	Conference and Workshop Papers	Emin Borandag
3	Journal Articles	Emin Borandag
4	Journal Articles	Emin Borandag
5	Journal Articles	Emin Borandag
6	Journal Articles	Emin Borandag
7	Journal Articles	Emin Borandag
8	Conference and Workshop Papers	Emin Borandag
9	Conference and Workshop Papers	Emin Borandag
10	Books and Theses	Emin Borandag

```
tony_df=get_data("Tony Fan")
tony_df["Name"]="Tony Fan"
tony_df
```

	Title	Year	\
0	Open X-Embodiment: Robotic Learning Datasets a...	2024	

1	A Critical Review on Hydrogen Production.	2023
2	A Review of 5G Building Management Technologie...	2023
3	A Critical Review on Smart Maintenance Decisio...	2022
4	Development of Wasted Non-woven Fabric Mask (N...	2022
5	Quantile-based fecal hemoglobin concentration ...	2019
6	Editorial TVLSI Positioning - Continuing and A...	2019
7	A Level Set Based Variational Principal Flow M...	2017
8	New region force for variational models in ima...	2017
9	Guarantees of Riemannian Optimization for Low ...	2016
10	Normal Estimation of a Transparent Object Usin...	2015
11	Energy Minimization Methods in Computer Vision...	2015
12	Mumford and Shah Model and Its Applications to...	2015
13	Numerical Methods and Applications in Total Va...	2015
14	Multi-class Transductive Learning Based on ℓ_1 ...	2014
15	Image Segmentation Using Euler's Elastica...	2013
16	A Ridge and Corner Preserving Model for Surfac...	2013
17	A Simple Compressive Sensing Algorithm for Par...	2013
18	Outdoor photometric stereo.	2013
19	Completely Convex Formulation of the Chan-Vese...	2012
20	Optimization of Surface Registrations Using Be...	2012
21	Erratum to: Optimization of Surface Registrati...	2012
22	A Novel Sparsity Reconstruction Method from Po...	2012
23	Erratum to: A Novel Sparsity Reconstruction Me...	2012
24	Computing quasiconformal maps using an auxilia...	2012
25	Intrinsic Feature Extraction on Hippocampal Su...	2012
26	Image Denoising Using Mean Curvature of Image ...	2012
27	Brain Surface Conformal Parameterization With ...	2012
28	DressUp!: outfit synthesis through automatic o...	2012
29	A framework for intrinsic image processing on ...	2011

	Type	Name
0	Conference and Workshop Papers	Tony Fan
1	Conference and Workshop Papers	Tony Fan
2	Conference and Workshop Papers	Tony Fan
3	Conference and Workshop Papers	Tony Fan
4	Conference and Workshop Papers	Tony Fan
5	Journal Articles	Tony Fan
6	Journal Articles	Tony Fan
7	Journal Articles	Tony Fan
8	Informal and Other Publications	Tony Fan
9	Journal Articles	Tony Fan
10	Journal Articles	Tony Fan
11	Editorship	Tony Fan
12	Reference Works	Tony Fan
13	Reference Works	Tony Fan
14	Journal Articles	Tony Fan
15	Journal Articles	Tony Fan
16	Journal Articles	Tony Fan
17	Journal Articles	Tony Fan

18	Conference and Workshop Papers	Tony Fan
19	Journal Articles	Tony Fan
20	Journal Articles	Tony Fan
21	Journal Articles	Tony Fan
22	Journal Articles	Tony Fan
23	Journal Articles	Tony Fan
24	Journal Articles	Tony Fan
25	Journal Articles	Tony Fan
26	Journal Articles	Tony Fan
27	Journal Articles	Tony Fan
28	Journal Articles	Tony Fan
29	Journal Articles	Tony Fan

2. Visualizations per Author

First, I want to visualize the data of each author separately

2.1 Building the visualization function

```
def get_visualizations(df, author_name):
    pubs_per_year = df.groupby(["Year", "Type"]).agg({"Title": "count"}).reset_index()

    pubs_per_type = df.groupby("Type").agg({"Title": "count"}).reset_index()
    bar = px.bar(pubs_per_year, x='Year', y='Title', title="{}'s publications per year".format(author_name), color="Type", barmode="group",
                 labels={"Year": "Year", "Title": "Number of Publications"})
    line = px.line(pubs_per_year, x="Year", y="Title", title="{}'s publications per year".format(author_name),
                  labels={"Year": "Year", "Title": "Number of Publications"})

    # pie = px.pie(pubs_per_type, values="Title", names="Type", title="Distribution of {}'s publications".format(author_name))

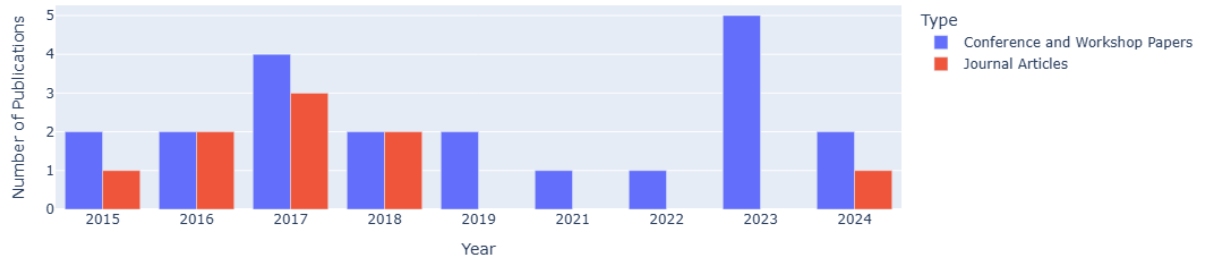
    pie = go.Figure(data=[go.Pie(labels=pubs_per_type["Type"], values=pubs_per_type["Title"], hole=0.3)])
    pie.update_layout(width=600, height=600, title="Distribution of {}'s publications".format(author_name))
    return bar, line, pie
```

2.2 Vasileios Tenentes

```
vas_bar, vas_line, vas_pie = get_visualizations(vas_df, "Vasileios Tenentes")
```

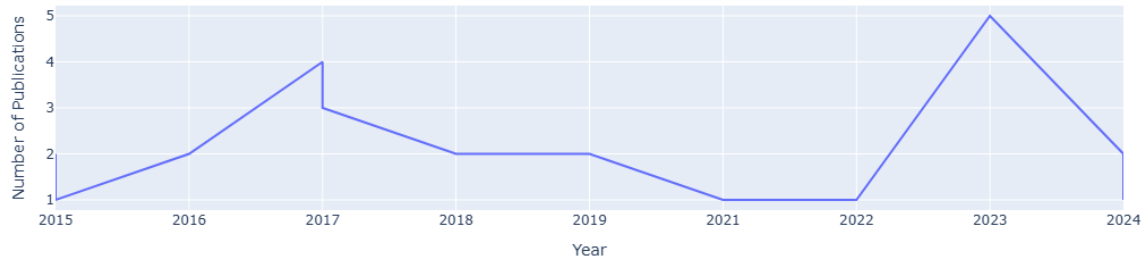
```
vas_bar.show()
```

Vasileios Tenentes's publications per year



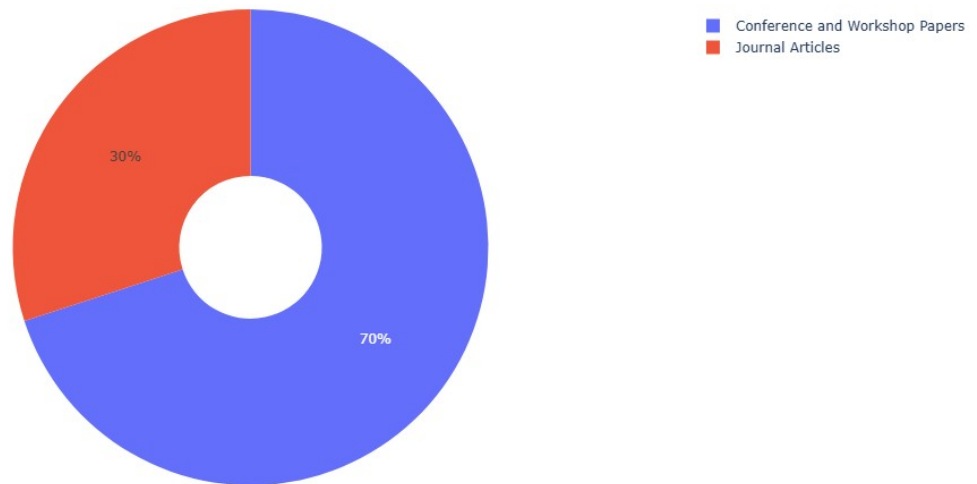
```
vas_line.show()
```

Vasileios Tenentes's publications per year



```
vas_pie.show()
```

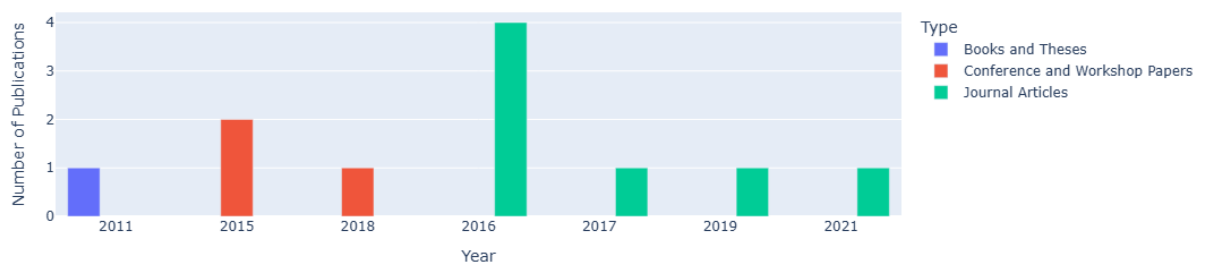
Distribution of Vasileios Tenentes's publications



2.3 Emin Borandag

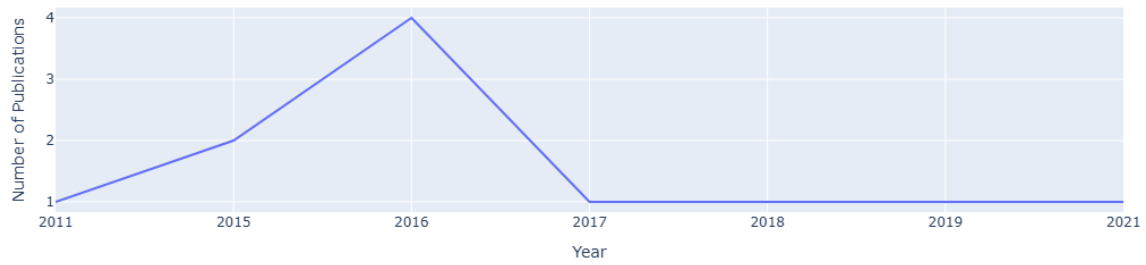
```
e_bar,e_line,e_pie=get_visualizations(emin_df,"Emin Borandag")  
e_bar.show()
```

Emin Borandag's publications per year



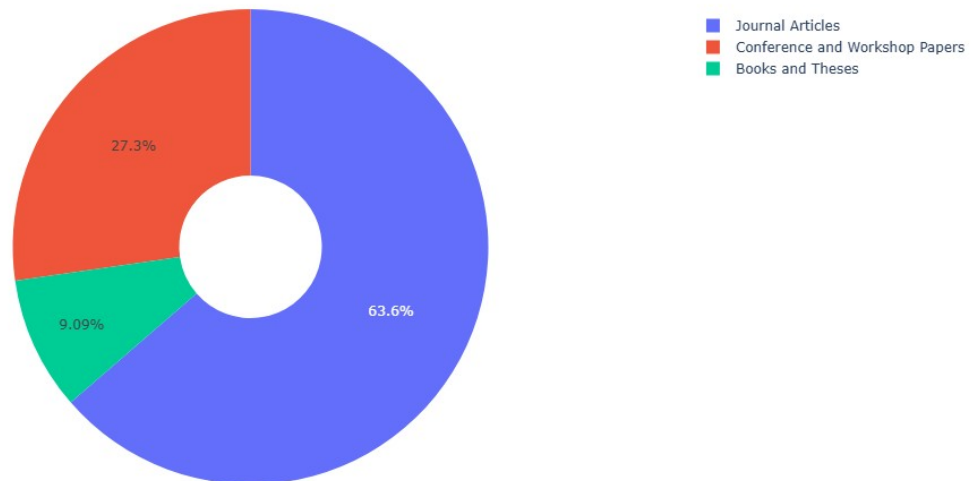
```
e_line.show()
```

Emin Borandag's publications per year



```
e_pie.show()
```

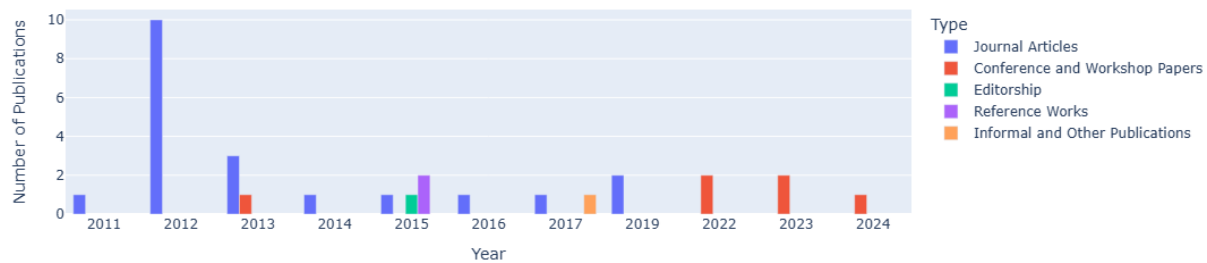
Distribution of Emin Borandag's publications



2.4 Tony Fan

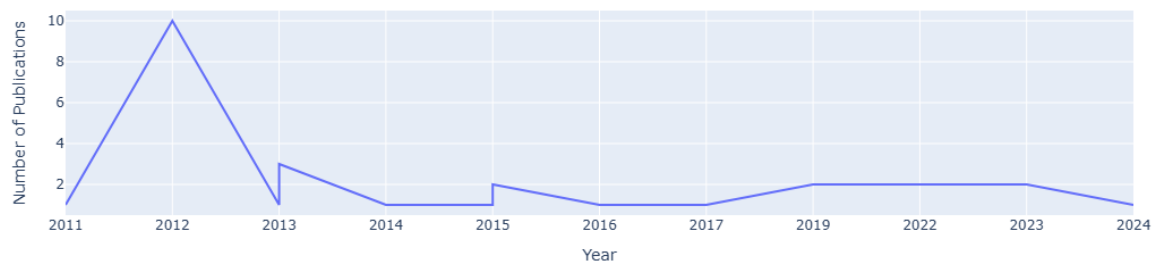
```
tony_bar,tony_line,tony_pie=get_visualizations(tony_df,"Tony Fan")  
tony_bar.show()
```


Tony Fan's publications per year



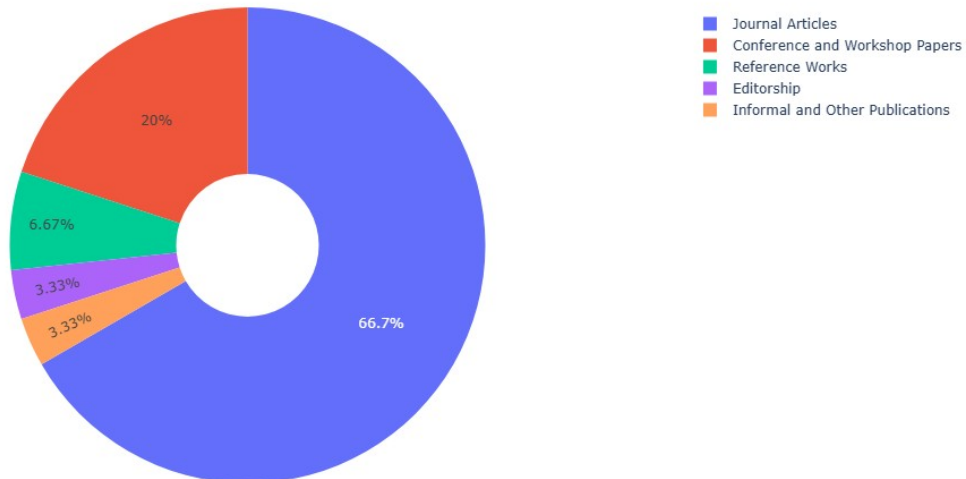
```
tony_line.show()
```

Tony Fan's publications per year



```
tony_pie.show()
```

Distribution of Tony Fan's publications



3.Cummulative visualizations

Creating a function that takes as arguments the three dataframes,merges them into one dataframe and returns 3 plots

```
def cumm_visualizations(df1,df2,df3):
    cumm_df=pd.concat([df1,df2,df3],axis=0).reset_index()

    pubs_per_year=cumm_df.groupby(["Name","Year"]).agg({"Title":"count"}).
    reset_index()

    pubs_per_type=cumm_df.groupby(["Type","Name"]).agg({"Title":"count"}).
    reset_index()
    bar= px.bar(pubs_per_year, x='Year',
    y='Title',color="Name",title="Distribution of the Authors'
    publications per year",category_orders=
        {"Year":
    sorted(pubs_per_year['Year'])},labels={"Title":"Number of
    Publications"},barmode="group")

    line=px.line(pubs_per_year,x="Year",y="Title",color="Name",title="Evol
    ution of the Authors' publications per year",category_orders=
        {"Year":sorted(pubs_per_year["Year"])})

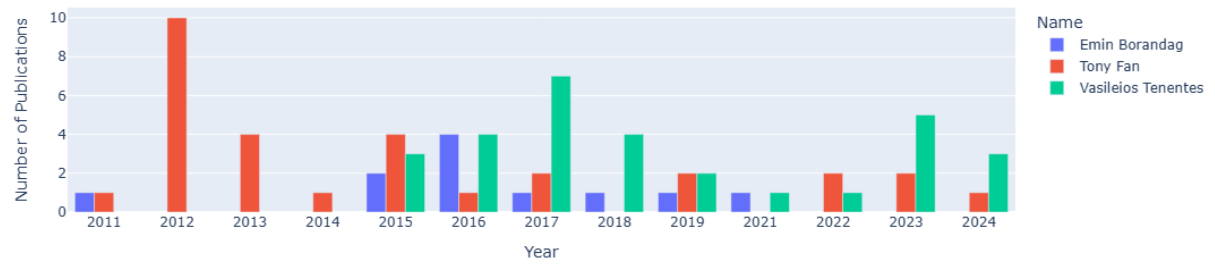
    #pie=px.pie(pubs_per_type,values="Title",names="Type",title="Distribut
    ion of the Authors' publication types")

    pie=go.Figure(data=[go.Pie(labels=pubs_per_type["Type"],values=pubs_pe
    r_type["Title"],hole=0.3)])
    pie.update_layout(width=700,height=700,title="Distribution of the
    cummulative publications")
    return bar,line,pie

cumm_bar,cumm_line,cumm_pie=cumm_visualizations(vas_df,emin_df,tony_df
)

cumm_bar.show()
```

Distribution of the Authors' publications per year



```
cumm_line.show()
```

Evolution of the Authors' publications per year



```
cumm_pie.show()
```

Distribution of the cummulative publications

